

SWE 4504

Writing a Basic Sniffer and Decoding IP Header

Task 1 (Basic Sniffer)

1. First make a virtual environment with python 3.11.9 (Not Mandatory, its my version so it advisable to follow my version) in the first terminal. Refer to Lab 4 Group Odd's slide to know about how to set up a virtual environment.
2. Write a basic sniffer in sniffer.py utilizing slide 7 or the book. **It will capture only one packet.**
3. Execute sniffer.py in the first terminal with : `sudo python sniffer.py`
4. Open second terminal. Ping nostarch.com using : `ping nostarch.com`
5. The output in the first terminal should be like the following :

```
(b'Ep\x00T\xa3\xb4\x00\x006\x01\x7f2h\x14\x11y\xc0\xa8'\x1d\x00\x00r\xe02\x13\x00\x01\xe9\xbf\xb6g\x00\x00\x00\x00\xf2\x10\n\x00\x00\x00\x00\x00\x10\x11\x12\x13\x14\x15\x16\x17\x18\x19\x1a\x1b\x1c\x1d\x1e\x1f\n!\"#$%&'\()*+,-./01234567', ('104.20.17.121', 0))
```

Basic Error Handling:

1. Make sure you have internet connection.
2. Make sure you are using your IP address in the HOST variable.

Task 2 (Bit More Complex Sniffer)

1. Open the first terminal and activate the same virtual environment that you used in task 1.
2. Write a sniffer in `sniffer_ip_header_decode.py` file. Make sure it can decode the header and show you the protocol, source and destination address.
3. Execute `sniffer_ip_header_decode.py` in the first terminal with : `sudo python sniffer_ip_header_decode.py`
4. Open the second terminal and ping google with : `ping google.com`
5. You should get the following output in the first terminal :

```
Protocol: ICMP 74.125.226.78 -> 192.168.0.190
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```

Basic Error handling:

1. Follow the task 1's basic error handling.